

Improving RSA Cryptanalysis: Combining Continued Fractions and Coppersmith's Techniques^{*}

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Abstract. In this paper, we present a new small private exponent attack on RSA by combining continued fractions and Coppersmith's techniques. Our results improve upon previous bounds, including Herrmann-May's attack, by leveraging a crucial relation derived from continued fraction. Additionally, we extend the range of vulnerable small private exponents by considering the partial leakage of prime factors or their sum. Our main result establishes an improved attack bound $d < N^{1-\alpha/3-\gamma/2}$, where $\alpha := \log_N e$ and $\gamma := \log_N |p + q - S|$, with S being an approximation of the prime sum $p + q$. Furthermore, we explore more applications of our main attack in scenarios where the primes share some most or least significant bits. The validity of our proposed main attack is confirmed through numerical experiments, demonstrating its improved performance over existing attacks.

Keywords: Attack · Continued fractions · Coppersmith's techniques · Factorization · RSA

1 Introduction

The RSA cryptosystem, originally introduced in 1978 by Rivest, Shamir, and Adleman [22], remains one of the most widely adopted public-key encryption schemes for secure communications. Despite its inception over four decades ago, RSA continues to stimulate research, particularly in the area of public-key cryptanalysis. In its simplified form, RSA uses the following parameters. Two large primes of the same size, p and q , are selected to form the modulus $N = pq$. Two integers, the public exponent e and the private exponent d , are then chosen to

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satisfy the congruence $ed \equiv 1 \pmod{\phi(N)}$, where $\phi(N) = (p-1)(q-1)$ denotes Euler's totient function.

Numerous attacks on RSA have been studied under various scenarios. One prominent line of research focuses on the vulnerabilities introduced by a small private exponent. It is a critical consideration for reducing decryption costs in RSA. In 1990, Wiener [32] demonstrated a key-recovery attack using continued fractions and Legendre's theorem, which is effective when $d < \frac{1}{3}N^{1/4}$. This bound was subsequently improved to $d < \frac{1}{\sqrt[4]{18}}N^{1/4}$ in [25]. Furthermore, Verheul et al. [30] enhanced Wiener's result by employing an exhaustive search over approximately $8 + 2c$ bits to obtain $d < 2^c N^{1/4}$, while Dujella [6] later proposed a similar exponential-time attack with additional refinements.

Other variants of Wiener's attack have been developed, though many require more than the standard public information (N, e) . For instance, de Weger's attack [31] exploits the small difference $|p - q|$ between the prime factors, whereas the attack by Blömer and May [2] assumes a linear relation of the form $ex + y \equiv 0 \pmod{\phi(N)}$, with x and y bounded. More recently, Susilo et al. [26] revisited the continued fraction-based approach by incorporating a strengthened version of Legendre's theorem.

Beyond continued fraction-based attacks, Coppersmith [5] pioneered lattice-based attacks on RSA, thereby opening new avenues in lattice-based cryptanalysis. Boneh and Durfee [3] advanced this approach to achieve an attack bound of $d < N^{0.292}$ for small private exponents. In 2010, Herrmann and May [11] further refined this method to obtain the same bound through a more efficient technique. Despite subsequent improvements, as seen in [27, 16, 8], the bound $d < N^{0.292}$ remains the state-of-the-art in practice.

Notably, these bounds can be further improved when partial leakage of the private key occurs. The concept of partial key exposure attacks was introduced by Boneh et al. [4] to address scenarios in which an adversary gains access to certain bits of the private exponent d . Building on this, Ernst et al. [7] demonstrated that knowledge of the most significant bits (MSBs) of d can facilitate more attacks even when d is relatively large. Similar techniques have been extended to cases involving the leakage of the prime factors p or q , or their sum $p + q$ [23, 21, 28, 8].

In practice, the attacks for small private exponents proposed in [3, 11] do not achieve these theoretical limits. Achieving the asymptotic bound would require operating on lattices of much higher dimensions, for which current lattice basis reduction algorithms struggle to provide both efficiency and quality. Consequently, experimental bounds are invariably lower than those predicted theoretically.

To bridge this gap, exhaustive search techniques are often employed. By guessing portions of the prime factors of the RSA modulus, it is possible to slightly raise the practical upper bound on the private exponent beyond $d < N^{0.292}$. For example, Li et al. [16] presented a practical attack that refines Herrmann-May's attack through a binary search for the MSBs of one prime. Subsequently, Feng et al. [8] proposed an improved attack that performs well

when the private exponent nears the theoretical limit using an exhaustive search over the MSBs of $p + q$.

Our Contribution. In this work, we present an enhanced analysis of small private exponent attacks on RSA by integrating continued fractions with Coppersmith’s lattice-based method. The literature predominantly employs two distinct approaches for small private exponent attacks: the continued fraction-based method (effective when $d < N^{0.25}$) and the more powerful lattice-based method (extending the bound to $d < N^{0.292}$). Although several hybrid attacks combining these methods have been proposed, our investigation seeks to determine whether a novel, distinct strategy can further improve the attack.

We begin by revisiting the continued fraction-based approach, thoroughly examining the properties of the convergents of $\frac{e}{N}$ and $\frac{e}{\phi(N)}$ to identify a common convergent $\frac{a}{b}$. Note that in RSA, the exponents e , d and Euler’s totient function $\phi(N)$ are related by $ed - k\phi(N) = 1$. This analysis not only clarifies the conditions under which the ratio $\frac{k}{d}$ appears among the convergents of $\frac{e}{N}$ but also shows the optimality of Wiener’s attack when solely relying on continued fraction.

Subsequently, we leverage the identified common convergent $\frac{a}{b}$ to enhance small private exponent attacks, particularly in scenarios where partial information like known MSBs of $p+q$ or shared MSBs/LSBs (where LSBs stands for least significant bits) of the primes is available. By expressing the RSA key equation $ed - k\phi(N) = 1$ in terms of parameters derived from the common convergents $\frac{a}{b}$ of $\frac{e}{N}$ and $\frac{e}{\phi(N)}$, we employ a lattice-based strategy to extract the roots of the resulting trivariate modular equations. Compared to Herrmann-May’s approach, which approximates $\phi(N)$, our method yields improved results. We also extend our main approach to attack scenarios with known MSBs/LSBs of the prime factors, achieving superior attack performance relative to previous work.

Organization. The remainder of this paper is organized as follows. In Section 2, we introduce the essential preliminaries, including both the continued fractions and lattice-based solving strategy. Section 3 revisits existing continued fraction-based attacks and examines its new properties. We also revisit lattice-based attacks that utilize approximations of the prime sum. In Section 4, we detail our new attack and its applications to scenarios where the prime factors share specific most or least significant bits. Experimental results validating our approach are presented in Section 5. Finally, we conclude the paper in Section 6.

2 Preliminaries

In this section, we introduce the fundamental concepts that support our new attacks. In particular, we recall the LLL lattice reduction algorithm introduced by Lenstra et al. [15] and Coppersmith’s lattice-based method [5] (later refined via Howgrave-Graham’s lemma [12]). For additional details, readers may refer to [18, 33]. We also summarize several well-known properties of continued fractions as presented in [10].

2.1 Continued Fractions

Let ξ be a nonzero real number. Its continued fraction expansion is given by

$$\xi = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{\ddots}}}},$$

where $a_0 \in \mathbb{Z}$ and $a_1, a_2, a_3, \dots$ are positive integers. This expansion is commonly abbreviated as $\xi = [a_0, a_1, a_2, a_3, \dots]$. The partial quotients a_i are determined by the recurrence

$$x_0 = \xi, \quad a_i = \lfloor x_i \rfloor, \quad x_{i+1} = \frac{1}{x_i - a_i}, \quad i \geq 0, \quad x_i - a_i \neq 0.$$

The n th convergent of ξ is the rational number $\frac{p_n}{q_n} = [a_0, a_1, \dots, a_n]$. According to the Euler-Wallis theorem, the convergents can be computed recursively:

$$\begin{aligned} p_0 &= a_0, & p_1 &= a_0 a_1 + 1, & p_n &= a_n p_{n-1} + p_{n-2}, & n &\geq 2, \\ q_0 &= 1, & q_1 &= a_1, & q_n &= a_n q_{n-1} + q_{n-2}, & n &\geq 2. \end{aligned}$$

A direct consequence is the identity $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n+1}$, which follows by induction. The convergents satisfy the inequality

$$\frac{p_0}{q_0} < \frac{p_2}{q_2} < \dots < \xi < \dots < \frac{p_3}{q_3} < \frac{p_1}{q_1}. \quad (1)$$

Moreover, for any integers k and d , there exists a unique solution (u, v) to the system

$$\begin{cases} k = up_r + vp_{r-1}, \\ d = uq_r + vq_{r-1}. \end{cases} \quad (2)$$

This solution is explicitly given by

$$u = \frac{kq_{r-1} - dp_{r-1}}{p_r q_{r-1} - q_r p_{r-1}}, \quad v = \frac{dp_r - kq_r}{p_r q_{r-1} - q_r p_{r-1}},$$

where the denominator equals $(-1)^{r+1}$, ensuring that u and v are integers. It is a crucial observation and will be further utilized in our main attack. A fundamental result due to Legendre [14] characterizes the convergents of ξ as follows.

Theorem 1 ([14]). *Let $\xi \in \mathbb{R}$ and let $a, b \in \mathbb{Z}$ with $\gcd(a, b) = 1$. If*

$$\left| \xi - \frac{a}{b} \right| < \frac{1}{2b^2},$$

then $\frac{a}{b}$ is a convergent of ξ .

2.2 Coppersmith's Lattice-Based Techniques

We review the lattice concepts essential to our attack. A lattice Λ is defined as the set of all integer linear combinations of linearly independent vectors $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_\omega \in \mathbb{R}^n$:

$$\Lambda = \left\{ \sum_{i=1}^{\omega} z_i \mathbf{b}_i : z_i \in \mathbb{Z} \right\}.$$

Its determinant, denoted $\det(\Lambda)$, is computed as $\sqrt{\det(BB^\top)}$, where the vectors $\mathbf{b}_i$ are the rows of the basis matrix B , and $B^\top$ is its transpose. For a full-rank lattice ($\omega = n$), this simplifies to $\det(\Lambda) = |\det(B)|$.

The LLL algorithm [15] is a pivotal tool for lattice reduction. It produces a reduced basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_\omega)$ satisfying good property.

Lemma 1 ([15, 18]). *For an ω -dimensional lattice Λ , the LLL algorithm computes a reduced basis $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_\omega)$ such that*

$$\|\mathbf{v}_1\| \leq \dots \leq \|\mathbf{v}_i\| \leq 2^{\frac{\omega(\omega-1)}{4(\omega+1-i)}} \det(\Lambda)^{\frac{1}{\omega+1-i}}, \quad i = 1, 2, \dots, \omega.$$

The algorithm runs in time polynomial in ω and the logarithm of the maximal input coefficient.

Howgrave-Graham [12] provided a criterion to lift a modular root to an integer root. For an integer polynomial $g(x_1, \dots, x_n) = \sum c_{i_1, \dots, i_n} x_1^{i_1} \dots x_n^{i_n}$, its norm is defined as $\|g(x_1, \dots, x_n)\| = \sqrt{\sum |c_{i_1, \dots, i_n}|^2}$.

Lemma 2 ([12]). *Let $g(x_1, \dots, x_n) \in \mathbb{Z}[x_1, \dots, x_n]$ be a polynomial consisting of at most ω monomials. Given positive integers $R, X_1, \dots, X_n$, suppose that*

$$g(x_1^*, \dots, x_n^*) \equiv 0 \pmod{R} \quad \text{and} \quad \|g(X_1 x_1, \dots, X_n x_n)\| < \frac{R}{\sqrt{\omega}},$$

where $|x_i^| \leq X_i$ for $i = 1, \dots, n$. Then $g(x_1^*, \dots, x_n^*) = 0$ over the integers.*

By combining the LLL algorithm with Howgrave-Graham's lemma, one can effectively solve modular (and subsequently integer) polynomial equations. Suppose we obtain the first ℓ reduced vectors $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_\ell)$ from the lattice. Success is ensured if this crucial solvable condition $2^{\frac{\omega(\omega-1)}{4(\omega+1-\ell)}} \det(\Lambda)^{\frac{1}{\omega+1-\ell}} < \frac{R}{\sqrt{\omega}}$ holds.

The lattice-based solving strategy has the following steps: ① Construct a set of shift polynomials from the given polynomial $f(x_1, \dots, x_n)$ using the estimated bounds $X_1, \dots, X_n$, ensuring they share a common root modulo R . ② Convert the coefficient vectors of the scaled shift polynomials $g_i(X_1 x_1, \dots, X_n x_n)$ into the rows of a lattice basis matrix. ③ Apply the LLL algorithm to obtain a reduced basis, from which one derives integer polynomials $h_i(x_1, \dots, x_n)$. ④ Provided the polynomials $h_i(x_1, \dots, x_n)$ are algebraically independent, solve the system (e.g., via resultants or Gröbner basis computation) to recover the common root $(x_1^*, \dots, x_n^*)$.

Although the construction of an optimal lattice basis has been extensively studied in [13, 17], we note that the lattice-based solving strategy is inherently

heuristic. There is no absolute guarantee that the derived polynomials will be algebraically independent. Nonetheless, it is standard practice in lattice-based attacks to assume their independence. Thus, we adopt the following assumption throughout this paper.

Assumption 1. *The integer polynomials derived from the LLL-reduced basis are algebraically independent and their common root can be efficiently solved.*

3 Revisiting Previous Attacks

We begin by reviewing several approximations concerning $\phi(N)$ that have been instrumental in previous attacks (see e.g., [19,20]).

Lemma 3 ([20]). *Let $N = pq$ be an RSA modulus with $q < p < 2q$, and $\phi(N) = (p-1)(q-1) = N+1-(p+q)$ the Euler's totient function. Then $2\sqrt{N} < p+q < \frac{3\sqrt{2}}{2}\sqrt{N}$, and*

$$N - \frac{3\sqrt{2}}{2}\sqrt{N} + 1 < \phi(N) < N - 2\sqrt{N} + 1.$$

We also recall approximations in scenarios where the primes p and q share either their most or least significant bits.

Lemma 4 ([31]). *Let $N = pq$ be an RSA modulus, then*

$$0 < p+q-2\sqrt{N} < \frac{(p-q)^2}{4\sqrt{N}}.$$

Lemma 5 ([20]). *Let $N = pq$ be an RSA modulus with $q < p < 2q$. Suppose that p and q share their least significant bits and $p-q = 2^n g$, where n is known and g is unknown. Let g_0 be a root of $x^2 \equiv N \pmod{2^n}$, and define $r_0 \equiv 2g_0 + (N - g_0^2)g_0^{-1} \pmod{2^{2n}}$. Then there exist positive integers p_0 , q_0 , and r_1 such that*

$$p = 2^n p_0 + g_0, \quad q = 2^n q_0 + g_0, \quad p+q = 2^{2n} r_1 + r_0.$$

3.1 Continued Fraction-Based Attacks

We present several results concerning the convergents derived from the continued fraction expansions involved with RSA parameters.

Lemma 6. *Let $N = pq$ be an RSA modulus with $\phi(N) = (p-1)(q-1)$. Let e be a public exponent and let d satisfy $ed \equiv 1 \pmod{\phi(N)}$. Then, there exists a positive integer k such that $ed - k\phi(N) = 1$, and*

- if $d < \frac{1}{2}\phi(N)$, then $\frac{k}{d}$ is a convergent of $\frac{e}{\phi(N)}$.
- if $\frac{1}{2}\phi(N) < d < \phi(N)$, then $\frac{e-k}{\phi(N)-d}$ is a convergent of $\frac{e}{\phi(N)}$.

Proof. Assume that $ed - k\phi(N) = 1$. If $d < \frac{1}{2}\phi(N)$, then

$$\left| \frac{e}{\phi(N)} - \frac{k}{d} \right| = \frac{1}{d\phi(N)} < \frac{1}{2d^2},$$

and, by Theorem 1, $\frac{k}{d}$ is a convergent of $\frac{e}{\phi(N)}$. If $\frac{1}{2}\phi(N) < d < \phi(N)$, then $\phi(N) - d < \frac{1}{2}\phi(N)$ and consequently, $\frac{1}{\phi(N)} < \frac{1}{2(\phi(N)-d)}$. Thus,

$$\left| \frac{e}{\phi(N)} - \frac{e-k}{\phi(N)-d} \right| = \frac{1}{\phi(N)(\phi(N)-d)} < \frac{1}{2(\phi(N)-d)^2}.$$

Since $e(\phi(N)-d) - \phi(N)(e-k) = -1$, we have $\gcd(e-k, \phi(N)-d) = 1$. Hence, by Theorem 1, $\frac{e-k}{\phi(N)-d}$ is a convergent of $\frac{e}{\phi(N)}$. $\square$

Theorem 2. Let $N = pq$ with $q < p < 2q$. Let e be a public exponent satisfying $\lfloor \frac{e}{N} \rfloor = \lfloor \frac{e}{\phi(N)} \rfloor = a_0$. If $\frac{a}{b}$ is a common convergent of both $\frac{e}{N}$ and $\frac{e}{\phi(N)}$, then

$$a < \frac{e - a_0 \left(N - \frac{3\sqrt{2}}{2} \sqrt{N} + 1 \right)}{\sqrt{e} N^{1/4}} \quad \text{and} \quad b < \frac{N^{3/4}}{\sqrt{e}}.$$

Proof. First, note that if $\phi(N) < e < N$, then $\frac{e}{N} < 1$ while $\frac{e}{\phi(N)} > 1$. Hence, they cannot share a common convergent. Then, suppose that either $e < \phi(N)$ or $e > N$ so that the two fractions have at least one common convergent. Write $\frac{e}{N} = [a_0, a_1, a_2, \dots]$ and $\frac{e}{\phi(N)} = [a_0, a'_1, a'_2, \dots]$, with $\lfloor \frac{e}{N} \rfloor = \lfloor \frac{e}{\phi(N)} \rfloor = a_0$. By Lemma 3, we have

$$\left| \frac{e}{\phi(N)} - \frac{e}{N} \right| = \frac{e(N - \phi(N))}{N\phi(N)} > \frac{2e\sqrt{N}}{N \left(N - \frac{3\sqrt{2}}{2} \sqrt{N} \right)} = \frac{2e}{N \left(\sqrt{N} - \frac{3\sqrt{2}}{2} \right)}.$$

If $\frac{a}{b}$ is a nonzero common convergent, then

$$\frac{2e}{N \left(\sqrt{N} - \frac{3\sqrt{2}}{2} \right)} \leq \left| \frac{e}{\phi(N)} - \frac{a}{b} \right| + \left| \frac{a}{b} - \frac{e}{N} \right| < \frac{2}{b^2},$$

which implies

$$b < \frac{\sqrt{N \left(\sqrt{N} - \frac{3\sqrt{2}}{2} \right)}}{\sqrt{e}} < \frac{N^{3/4}}{\sqrt{e}}.$$

Furthermore, note that the reciprocal $\frac{b}{a}$ is a convergent of both $\frac{N}{e-a_0N}$ and $\frac{\phi(N)}{e-a_0\phi(N)}$. Applying Lemma 3 again, we obtain

$$\frac{N}{e-a_0N} - \frac{\phi(N)}{e-a_0\phi(N)} = \frac{e(N - \phi(N))}{(e-a_0N)(e-a_0\phi(N))} > \frac{2e\sqrt{N}}{(e-a_0N)(e-a_0\phi(N))}.$$

Thus,

$$\frac{2e\sqrt{N}}{(e - a_0N)(e - a_0\phi(N))} < \left| \frac{N}{e - a_0N} - \frac{b}{a} \right| + \left| \frac{b}{a} - \frac{\phi(N)}{e - a_0\phi(N)} \right| < \frac{2}{a^2}.$$

Since $e - a_0N < e - a_0\phi(N) < e - a_0\left(N - \frac{3\sqrt{2}}{2}\sqrt{N} + 1\right)$, it follows that

$$a < \frac{\sqrt{(e - a_0N)(e - a_0\phi(N))}}{\sqrt{e} N^{1/4}} < \frac{e - a_0\left(N - \frac{3\sqrt{2}}{2}\sqrt{N} + 1\right)}{\sqrt{e} N^{1/4}}.$$

This terminates the proof. $\square$

Corollary 1. *Let a_0 be a positive integer. If $e \in (1, \sqrt{N}) \cup (a_0\phi(N), a_0N) \cup (N^{3/2}, +\infty)$, then $\frac{e}{\phi(N)}$ and $\frac{e}{N}$ have no nontrivial common convergents.*

Proof. We consider the following three cases:

1. If $e \in (1, \sqrt{N})$, then $a_0 = \lfloor e/N \rfloor = 0$, and Theorem 2 implies any nontrivial common convergent $\frac{a}{b}$ would satisfy $a < \frac{\sqrt{e}}{N^{1/4}} < 1$, which is impossible.
2. If a_0 is a positive integer, and $e \in (a_0\phi(N), a_0N)$, then, since $a_0 < \frac{e}{\phi(N)}$, then $a_0 \leq \lfloor \frac{e}{\phi(N)} \rfloor$. Hence, the inequalities (1) imply that all the convergents of $\frac{e}{\phi(N)}$ are greater than a_0 . On the other hand, since $\frac{e}{N} < a_0$, then $\lfloor \frac{e}{N} \rfloor \leq a_0 - 1$. Hence, all the convergents of $\frac{e}{N}$ are less than a_0 and the convergents of $\frac{e}{N}$ and $\frac{e}{\phi(N)}$ lie on opposite sides of a_0 , precluding any nontrivial common convergent.
3. If $e > N^{3/2}$, then Theorem 2 implies that any common nontrivial convergent $\frac{a}{b}$ must satisfy $b < \frac{N^{3/4}}{\sqrt{e}} < 1$, which is impossible. $\square$

Therefore, Corollary 1 implies that $\frac{e}{N}$ and $\frac{e}{\phi(N)}$ may share a nontrivial common convergent for various intervals of e . The following result identifies conditions under which the convergents of $\frac{e}{N}$ cannot be used to attack RSA.

Theorem 3. *Let $N = pq$ be an RSA modulus with $q < p < 2q$ and $\phi(N) = (p-1)(q-1)$. Suppose that e is a public exponent with $a_0 = \lfloor \frac{e}{N} \rfloor$. Let d satisfy $ed \equiv 1 \pmod{\phi(N)}$ with $ed - k\phi(N) = 1$. Then $\frac{k}{d}$ is not among the convergents of $\frac{e}{N}$ if either*

1. $\frac{N^{3/4}}{\sqrt{e}} < d < \frac{1}{2}\phi(N)$ or $k > \frac{e - a_0\left(N - \frac{3\sqrt{2}}{2}\sqrt{N} + 1\right)}{\sqrt{e} N^{1/4}}$, or
2. $\frac{1}{2}\phi(N) < d < \phi(N) - \frac{N^{3/4}}{\sqrt{e}}$ or $k < e - \frac{e - a_0\left(N - \frac{3\sqrt{2}}{2}\sqrt{N} + 1\right)}{\sqrt{e} N^{1/4}}$.

Proof. Assume $ed - k\phi(N) = 1$ with $d < \phi(N)$. By Lemma 6, if $d < \frac{1}{2}\phi(N)$, then $\frac{k}{d}$ is a convergent of $\frac{e}{\phi(N)}$, whereas if $\frac{1}{2}\phi(N) < d < \phi(N)$, then $\frac{e-k}{\phi(N)-d}$ is a convergent of $\frac{e}{\phi(N)}$. In both cases, take $a_0 = \lfloor \frac{e}{N} \rfloor$.

If $d < \frac{1}{2}\phi(N)$, Theorem 2 implies that $\frac{k}{d}$ is not a convergent of $\frac{e}{N}$ provided that

$$k > \frac{e - a_0 \left(N - \frac{3\sqrt{2}}{2} \sqrt{N} + 1 \right)}{\sqrt{e} N^{1/4}} \quad \text{or} \quad d > \frac{N^{3/4}}{\sqrt{e}}.$$

Similarly, if $\frac{1}{2}\phi(N) < d < \phi(N)$, then $\frac{e-k}{\phi(N)-d}$ is not a convergent of $\frac{e}{N}$ if

$$e - k > \frac{e - a_0 \left(N - \frac{3\sqrt{2}}{2} \sqrt{N} + 1 \right)}{\sqrt{e} N^{1/4}} \quad \text{or} \quad \phi(N) - d > \frac{N^{3/4}}{\sqrt{e}}.$$

Equivalently, these conditions can be expressed as

$$k < e - \frac{e - a_0 \left(N - \frac{3\sqrt{2}}{2} \sqrt{N} + 1 \right)}{\sqrt{e} N^{1/4}} \quad \text{or} \quad d < \phi(N) - \frac{N^{3/4}}{\sqrt{e}}.$$

This completes the proof. $\square$

The following result addresses the case when the public exponent e is of full size and demonstrates that Wiener's attack is nearly optimal.

Theorem 4. *Let $N = pq$ be an RSA modulus with $q < p < 2q$, and let $e < \phi(N)$ be a public exponent satisfying $e \approx N$. If d satisfies $ed \equiv 1 \pmod{\phi(N)}$, then the convergents of $\frac{e}{N}$ fail to recover d if $N^{1/4} < d < \phi(N) - N^{1/4}$.*

Proof. Because $e < \phi(N)$ and $e \approx N$, we have $a_0 = \lfloor e/N \rfloor = 0$. Then the first condition in Theorem 3 becomes $\frac{N^{3/4}}{\sqrt{e}} < d < \frac{1}{2}\phi(N)$, which implies $N^{1/4} < d < \frac{1}{2}\phi(N)$. The second condition in Theorem 3 yields $k > \frac{e}{\sqrt{e} N^{1/4}} \approx N^{1/4}$. Hence, we have

$$d = \frac{k\phi(N) + 1}{e} > \frac{k\phi(N)}{e} > k > N^{1/4}.$$

Similarly, the third condition requires $\frac{1}{2}\phi(N) < d < \phi(N) - \frac{N^{3/4}}{\sqrt{e}}$, i.e., $\frac{1}{2}\phi(N) < d < \phi(N) - N^{1/4}$. Finally, the fourth condition gives $k < e - \frac{e}{\sqrt{e} N^{1/4}} \approx e - N^{1/4}$, so that

$$d = \frac{k\phi(N) + 1}{e} < \frac{(e - N^{1/4})\phi(N)}{e} < e - N^{1/4} < \phi(N) - N^{1/4}.$$

Thus, the convergents of $\frac{e}{N}$ cannot yield d when $N^{1/4} < d < \phi(N) - N^{1/4}$. $\square$

3.2 Lattice-Based Attacks

We revisit Herrmann-May's attack [11] for solving the RSA key equation $ed - k(p-1)(q-1) = 1$, under the assumption that an approximation S of $p+q$ is known, with $|p+q-S| < N^\gamma$ for $1/4 < \gamma \leq 1/2$.

Theorem 5. *Let $N = pq$ be an RSA modulus with $q < p < 2q$ and let $e = N^\alpha$ be a public exponent satisfying $ed - k(p-1)(q-1) = 1$ with $d < N^{\delta_0}$. Assume that an approximation S of $p+q$ is known such that $|p+q-S| < N^\gamma$ for $1/4 < \gamma \leq 1/2$. Then N can be factored in polynomial time provided that*

$$\delta_0 < 1 - \sqrt{\alpha\gamma}.$$

Proof. Let x denote the unknown k and set $y = p+q-S$. The key equation $ed - k(p-1)(q-1) = 1$ can be rewritten as $x(S-N-1+y) - 1 \equiv 0 \pmod{e}$. Define the polynomial

$$f(x, y) = x(A+y) - 1, \quad \text{with } A = S - N - 1.$$

Then $(x_0, y_0) = (k, p+q-S)$ is a small solution to $f(x, y) \equiv 0 \pmod{e}$. To apply the lattice-based strategy in [11], we introduce an auxiliary variable $u = xy - 1$.

Let N^{δ_0} be a bound for d and N^γ a bound for $|p+q-S|$. Noting that $k < \frac{ed}{(p-1)(q-1)} \approx N^{\alpha+\delta_0-1}$, we set

$$X = N^{\alpha+\delta_0-1}, \quad Y = N^\gamma, \quad \text{and} \quad U \approx XY = N^{\alpha+\delta_0+\gamma-1}.$$

Let m and t be two positive integers. Construct a lattice $\mathcal{L}$ using the shift polynomials

$$g_{k,i,j}(x, y, u) = x^i y^j f(x, y)^k e^{m-k},$$

where each occurrence of $xy - 1$ is replaced by u , and $(k, i, j) \in I \cup J$ with

$$I = \{(k, i, j) : j = 0, k = 0, \dots, m, i = 0, \dots, m-k\}$$

and

$$J = \{(k, i, j) : i = 0, j = 1, \dots, t, k = \lfloor m/t \rfloor j, \dots, m\}.$$

The lattice basis is formed by the coefficient vectors of the shift polynomials $g_{k,i,j}(Xx, Yy, Uu)$ (ordered lexicographically by (k, i, j)). The monomials $x^i y^j u^k$ are ordered similarly. This construction yields a triangular lattice matrix whose determinant has the form

$$\det(\mathcal{L}) = X^{n_X} Y^{n_Y} U^{n_U} e^{n_e}.$$

The exponents are $n_X = S(i)$, $n_Y = S(j)$, $n_U = S(k)$, $n_e = S(m-k)$ determined by the sums of the respective degrees, and lattice dimension is $\omega = S(1)$ for

$$S(\sigma) = \sum_{k=0}^m \sum_{i=0}^{m-k} \sigma + \sum_{j=1}^t \sum_{k=\lfloor m/t \rfloor j}^m \sigma.$$

For simplicity, we set $t = m\tau$ with $\tau > 0$ and approximate $\lfloor m/t \rfloor j \approx j/\tau$. Then, omitting lower-order terms, we obtain $\omega \approx \frac{1}{2}(\tau+1)m^2$ and

$$n_X \approx \frac{1}{6}m^3, \quad n_Y \approx \frac{1}{6}\tau^2 m^3, \quad n_U \approx \frac{1}{6}(2\tau+1)^2 m^3, \quad n_e \approx \frac{1}{6}(\tau+2)m^3.$$

By applying the LLL algorithm (Lemma 1) with $i = 3$ and then invoking Howgrave-Graham's lemma (Lemma 2), one deduces that

$$2^{\frac{\omega(\omega-1)}{4(\omega-2)}} \det(\mathcal{L})^{\frac{1}{\omega-2}} < \frac{e^m}{\sqrt{\omega}}.$$

After simplification, this condition reduces to

$$\gamma\tau^2 + 2(\delta_0 + \gamma - 1)\tau + \gamma + \alpha + 2\delta_0 - 2 < 0.$$

The left term is optimized by setting τ to be $\frac{1-\delta_0-\gamma}{\gamma}$, leading to

$$-\delta_0^2 + 2\delta_0 + \alpha\gamma - 1 < 0,$$

which in turn implies

$$\delta_0 < 1 - \sqrt{\alpha\gamma}.$$

□

4 Combining Continued Fractions and Coppersmith's Techniques

In this section, we integrate the relation (2) obtained from continued fractions into lattice-based attacks on small private exponents. In particular, we present new attacks that exploit both the information from (2) and the partial leakage of prime factors.

4.1 The Main Attack

Theorem 6. *Let $N = pq$ be an RSA modulus with $q < p < 2q$, and let $e = N^\alpha$ be a public exponent satisfying $ed - k(p-1)(q-1) = 1$. Let $\frac{p_{r-1}}{q_{r-1}}$ and $\frac{p_r}{q_r}$ be two consecutive convergents of $\frac{e}{N}$ with the largest denominators subject to $q_{r-1} < q_r < \frac{N^{3/4}}{\sqrt{e}}$ and $\gcd(p_r, e) = 1$. Assume that there exist integers u and v with $|u|, |v| < N^\delta$, such that $d = uq_r + vq_{r-1}$ and $k = up_r + vp_{r-1}$. Further, let S be an approximation of $p + q$ satisfying $|p + q - S| < N^\gamma$ for $1/4 < \gamma \leq 1/2$. Then, N can be factored in polynomial time provided that*

$$\delta < \frac{1}{6}\alpha - \frac{1}{2}\gamma + \frac{1}{4}.$$

Proof. Let $\frac{p_{r-1}}{q_{r-1}}$ and $\frac{p_r}{q_r}$ be two consecutive convergents of $\frac{e}{N}$ satisfying $q_{r-1} < q_r < \frac{N^{3/4}}{\sqrt{e}}$ and $\gcd(e, p_r) = 1$. Assume that $ed - k(p-1)(q-1) = 1$ with $d = uq_r + vq_{r-1}$ and $k = up_r + vp_{r-1}$ and that $p + q = S + w$, where S is known and w is unknown. Then the key equation can be rewritten as

$$e(uq_r + vq_{r-1}) - (up_r + vp_{r-1})(N + 1 - S - w) = 1.$$

Rearranging terms, we obtain

$$p_r w u + p_{r-1} w v - (N+1-S)p_r u - \left((N+1-S)p_{r-1} - eq_{r-1}\right)v - 1 \equiv 0 \pmod{eq_r}.$$

Multiplying by the inverse $p_r^{-1} \pmod{eq_r}$, we deduce

$$w u + a_1 w v + a_2 u + a_3 v + a_4 \equiv 0 \pmod{eq_r},$$

where

$$\begin{aligned} a_1 &\equiv p_{r-1} p_r^{-1} \pmod{eq_r}, \\ a_2 &\equiv -(N+1-S) \pmod{eq_r}, \\ a_3 &\equiv -\left((N+1-S)p_{r-1} - eq_{r-1}\right) p_r^{-1} \pmod{eq_r}, \\ a_4 &\equiv -p_r^{-1} \pmod{eq_r}. \end{aligned}$$

We define the polynomial

$$f(x, y, z) = xy + a_1 xz + a_2 y + a_3 z + a_4.$$

Then the triple $(x_0, y_0, z_0) = (w, u, v)$ is a solution of the congruence $f(x, y, z) \equiv 0 \pmod{eq_r}$. We apply the lattice-based strategy of Jochemsz-May [13]. Let m and t be two positive integers. Consider the monomial set for $0 \leq k \leq m$:

$$M_k = \bigcup_{0 \leq h \leq t} \left\{ x^i y^j z^{l+h} : x^i y^j z^l \text{ is a monomial in } f^m(x, y, z), \right. \\ \left. \text{and } \frac{x^i y^j z^l}{(xy)^k} \text{ is a monomial in } f^{m-k}(x, y, z) \right\}.$$

An analysis shows that the monomials in $f^m(x, y, z)$ are of the form

$$x^i y^j z^l : i = 0, \dots, m, \quad j = 0, \dots, m, \quad l = \max(i-j, 0), \dots, m-j,$$

while those in $f^{m-k}(x, y, z)$ satisfy

$$i = 0, \dots, m-k, \quad j = 0, \dots, m-k, \quad l = \max(i-j, 0), \dots, m-k-j.$$

Hence, the monomials in M_k can be written as

$$x^{i-k} y^{j-k} z^l : i = k, \dots, m, \quad j = k, \dots, m, \quad l = \max(i-j, 0), \dots, m-j+t.$$

Similarly, the monomials in M_{k+1} are those with

$$i = k+1, \dots, m, \quad j = k+1, \dots, m, \quad l = \max(i-j, 0), \dots, m-j+t.$$

The Jochemsz-May strategy employs the set difference $M_k \setminus M_{k+1}$, which consists of monomials of the form $x^{i-k} y^{j-k} z^l$ for indices (i, j, l) in the sets

$$I_k = \{(i, j, l) : i = k, j = k, \dots, m, l = \max(i-j, 0), \dots, m-j+t\},$$

$$J_k = \{(i, j, l) : i = k+1, \dots, m, j = k, l = \max(i-j, 0), \dots, m-j+t\}.$$

For each $(i, j, l) \in I_k \cup J_k$, define the shift polynomial

$$g_{k,i,j,l}(x, y, z) = x^{i-k} y^{j-k} z^l f^k(x, y, z) (eq_r)^{m-k}.$$

Then for every $0 \leq k \leq m$ and every $(i, j, l) \in I_k \cup J_k$, we have

$$g_{k,i,j,l}(x_0, y_0, z_0) \equiv 0 \pmod{eq_r}.$$

Let X, Y , and Z be bounds for $x_0 = w$, $y_0 = u$, and $z_0 = v$, respectively. We set

$$|x_0| < X = N^\gamma, \quad |y_0| < Y = N^\delta, \quad |z_0| < Z = N^\delta.$$

Using the coefficient vectors of the scaled polynomials $g_{k,i,j,l}(Xx, Yy, Zz)$, we construct a lattice $\mathcal{L}$. We order the rows of the lattice matrix lexicographically by (k, i, j, l) and the columns corresponding to the monomials $x^i y^j z^l$ similarly. With this ordering, the lattice matrix is triangular, and its determinant takes the form

$$\det(\mathcal{L}) = X^{n_X} Y^{n_Y} Z^{n_Z} (eq_r)^{n_e}.$$

The exponents $n_X = S(i)$, $n_Y = S(j)$, $n_Z = S(l)$, $n_e = S(m-k)$ and the lattice dimension $\omega = S(1)$ are defined via appropriate summations $S(\sigma)$ with

$$S(\sigma) = \sum_{k=0}^m \sum_{i=k}^k \sum_{j=k}^m \sum_{l=0}^{m-j+t} \sigma + \sum_{k=0}^m \sum_{i=k+1}^m \sum_{j=k}^k \sum_{l=i-j}^{m-j+t} \sigma.$$

In particular, setting $t = m\tau$ with $\tau > 0$ for simplicity, we obtain

$$\begin{aligned} n_X &= \frac{1}{8}(4\tau + 1)m^4 + o(m^4), \\ n_Y &= \frac{1}{8}(4\tau + 1)m^4 + o(m^4), \\ n_Z &= \frac{1}{8}(2\tau + 1)^2 m^4 + o(m^4), \\ n_e &= \frac{1}{12}(8\tau + 3)m^4 + o(m^4), \\ \omega &= \frac{1}{3}(3\tau + 1)m^3 + o(m^3). \end{aligned} \tag{3}$$

Using $e = N^\alpha$ and $q_r < \frac{N^{3/4}}{\sqrt{e}}$, we get $eq_r < \frac{eN^{3/4}}{\sqrt{e}} = N^{\frac{\alpha}{2} + \frac{3}{4}}$. To mix Lemma 1 with $i = 3$, and Lemma 2, we set

$$2^{\frac{\omega(\omega-1)}{4(\omega-2)}} \det(\mathcal{L})^{\frac{1}{\omega-2}} < \frac{(eq_r)^m}{\sqrt{\omega}}.$$

Extracting the exponents from the solvable condition

$$X^{n_X} Y^{n_Y} Z^{n_Z} (eq_r)^{n_e} < \frac{1}{2^{\frac{\omega(\omega-1)}{4}} \omega^{\frac{\omega-2}{2}}} (eq_r)^{(\omega-2)m},$$

and substituting the above estimates (3), we deduce after simplification that

$$24\delta\tau^2 + 4(12\delta - 2\alpha + 6\gamma - 3)\tau + 12\delta - 2\alpha + 6\gamma - 3 < 0.$$

The quadratic expression in τ attains its minimum at

$$\frac{2\alpha - 12\delta - 6\gamma + 3}{12\delta}.$$

After further simplification, it yields the condition

$$\delta < \frac{1}{6}\alpha - \frac{1}{2}\gamma + \frac{1}{4}. \quad (4)$$

This completes the proof. $\square$

For comparison, Herrmann-May's attack via Theorem 5 gives the bound $\delta_0 < 1 - \sqrt{\alpha\gamma}$. In our new attack, since $d = q_r u + q_{r-1} v$ with $q_{r-1} < q_r < N^{\frac{3}{4} - \frac{\alpha}{2}}$, and $|u|, |v| < N^\delta$ with δ satisfying (4), the overall bound on $d < N^{\delta_0}$ is given by

$$\delta_0 = \delta + \frac{3}{4} - \frac{\alpha}{2} < 1 - \frac{1}{3}\alpha - \frac{1}{2}\gamma.$$

Furthermore, the difference between our bound and that of Theorem 5 is

$$\begin{aligned} \Delta &= 1 - \frac{1}{3}\alpha - \frac{1}{2}\gamma - (1 - \sqrt{\alpha\gamma}) = \sqrt{\alpha\gamma} - \left(\frac{1}{3}\alpha + \frac{1}{2}\gamma\right) \\ &= -\frac{\gamma}{3} \left(\sqrt{\frac{\alpha}{\gamma}} - \frac{3 + \sqrt{3}}{2} \right) \left(\sqrt{\frac{\alpha}{\gamma}} - \frac{3 - \sqrt{3}}{2} \right). \end{aligned}$$

Hence $\Delta > 0$ whenever $\frac{6-3\sqrt{3}}{2}\gamma < \alpha < \frac{6+3\sqrt{3}}{2}\gamma$. In most cases, we have $\gamma \approx \frac{1}{2}$, leading to $\frac{6-3\sqrt{3}}{4} < \alpha < \frac{6+3\sqrt{3}}{4}$, namely $0.201 < \alpha < 2.799$ that is the most used interval for e . It follows that, in most choices of e , we have $\Delta > 0$, implying that our improved attack yields a better theoretical bound than that derived by using Herrmann-May's attack.

Remark 1. It is worth noting that the lattice construction used in the proof of Theorem 6 might be further optimized. For example, by using an additional shift on the variable x , one might obtain an improved bound in certain regions for smaller γ with a fixed α . However, such modifications introduce heavy analytical complexity and tiny resulting improvement in the bound. Both this asymptotic bound and the one presented earlier in our main attack can also be computed using the formulas provided in [9] and their code[‡].

[‡] <https://github.com/ffmath/AsymptoticBounds>

4.2 Applications to Primes Sharing MSBs or LSBs

We present two more applications of the main attack in scenarios where the primes share certain MSBs or LSBs.

Theorem 7. *Let $N = pq$ be an RSA modulus with $q < p < 2q$, and let $e = N^\alpha$ be a public exponent satisfying $ed - k(p-1)(q-1) = 1$. Let $\frac{p_{r-1}}{q_{r-1}}$ and $\frac{p_r}{q_r}$ be two consecutive convergents of $\frac{e}{N}$ with $q_{r-1} < q_r < \frac{N^{3/4}}{\sqrt{e}}$ and $\gcd(e, p_r) = 1$. Suppose there exist integers u and v with $|u|, |v| < N^\delta$ such that $d = uq_r + vq_{r-1}$ and $k = up_r + vp_{r-1}$, and assume that p and q share their most significant bits so that $p - q < N^\beta$, with $1/4 < \beta \leq 1/2$. Then, N can be factored in polynomial time provided that*

$$\delta < \frac{1}{6}\alpha - \beta + \frac{1}{2}.$$

Proof. Assume that $p - q = N^\beta$. Then by Lemma 4 we have

$$p + q - 2\sqrt{N} < \frac{(p - q)^2}{4\sqrt{N}} = \frac{1}{4}N^{2\beta - \frac{1}{2}}.$$

Thus, writing $p + q = S + w$ with $S = \lfloor 2\sqrt{N} \rfloor$ known and $w < \frac{1}{4}N^{2\beta - \frac{1}{2}}$, we set

$$X = \frac{1}{4}N^{2\beta - \frac{1}{2}},$$

so that $\gamma = 2\beta - \frac{1}{2}$. Then the condition (4) becomes

$$\delta < \frac{1}{6}\alpha - \frac{1}{2}\left(2\beta - \frac{1}{2}\right) + \frac{1}{4} = \frac{1}{6}\alpha - \beta + \frac{1}{2}.$$

This completes the proof. $\square$

For comparison, if we set $S = 2\sqrt{N}$ so that $|p + q - S| < N^\gamma$ with $\gamma = 2\beta - \frac{1}{2}$, Theorem 5 yields the bound $\delta_0 < 1 - \sqrt{(2\beta - \frac{1}{2})\alpha}$. In our new attack via Theorem 7, we obtain an overall bound on $d < N^{\delta_0}$ with

$$\delta_0 = \delta + \frac{3}{4} - \frac{\alpha}{2} < \frac{5}{4} - \frac{1}{3}\alpha - \beta.$$

Thus, a similar analysis shows that our new bound is superior than the bound in Theorem 5 (i.e., it is similar to the comparison in Section 4.1 using $\gamma = 2\beta - \frac{1}{2}$).

Next, we consider another attack case when p and q share some of their least significant bits.

Theorem 8. *Let $N = pq$ be an RSA modulus with $q < p < 2q$, and let $e = N^\alpha$ be a public exponent satisfying $ed - k(p-1)(q-1) = 1$. Let $\frac{p_{r-1}}{q_{r-1}}$ and $\frac{p_r}{q_r}$ be two consecutive convergents of $\frac{e}{N}$ with $q_{r-1} < q_r < \frac{N^{3/4}}{\sqrt{e}}$ and $\gcd(e, p_r) = 1$. Suppose there exist integers u and v with $|u|, |v| < N^\delta$ such that $d = uq_r + vq_{r-1}$ and*

$k = up_r + vp_{r-1}$, and assume that p and q share their least significant bits so that $p - q = 2^n g$ with $2^n = N^\beta$, $\beta < \frac{1}{4}$, where n is known and g is unknown. Then, N can be factored in polynomial time provided that

$$\delta < \frac{1}{6}\alpha + \beta.$$

Proof. Assume that $p - q = 2^n g$ with known n and unknown g . By Lemma 5, let g_0 be a solution of $x^2 \equiv N \pmod{2^n}$, and define $r_0 \equiv 2g_0 + (N - g_0^2)g_0^{-1} \pmod{2^{2n}}$. Then we may express $p + q = 2^{2n}r_1 + r_0$, for some unknown integer r_1 . The key equation $ed - k(p-1)(q-1) = 1$ can be rewritten as

$$e(uq_r + vq_{r-1}) - (up_r + vp_{r-1})(N + 1 - 2^{2n}r_1 - r_0) = 1.$$

Rearranging terms yields

$$\begin{aligned} 2^{2n}p_r r_1 u + 2^{2n}p_{r-1} r_1 v - p_r(N + 1 - r_0)u \\ - \left((N + 1 - r_0)p_{r-1} - eq_{r-1} \right)v - 1 \equiv 0 \pmod{eq_r}. \end{aligned}$$

Assuming $\gcd(2p_r, eq_r) = 1$, we multiply by $2^{-2n}p_r^{-1} \pmod{eq_r}$ to obtain

$$r_1 u + a_1 r_1 v + a_2 u + a_3 v + a_4 \equiv 0 \pmod{eq_r},$$

where

$$\begin{aligned} a_1 &\equiv p_{r-1}p_r^{-1} \pmod{eq_r}, \\ a_2 &\equiv -2^{-2n}(N + 1 - r_0) \pmod{eq_r}, \\ a_3 &\equiv -2^{-2n}\left((N + 1 - r_0)p_{r-1} - eq_{r-1}\right)p_r^{-1} \pmod{eq_r}, \\ a_4 &\equiv -2^{-2n}p_r^{-1} \pmod{eq_r}. \end{aligned}$$

We define the polynomial

$$f(x, y, z) = xy + a_1xz + a_2y + a_3z + a_4.$$

Then the triple (r_1, u, v) is a solution of $f(x, y, z) \equiv 0 \pmod{eq_r}$. Following the same strategy as in the proof of Theorem 6 (with r_1 in place of w), and setting $2^n = N^\beta$, we have

$$r_1 = \frac{p + q - r_0}{2^{2n}} < \frac{p + q}{N^{2\beta}} < 3N^{\frac{1}{2} - 2\beta}.$$

Taking $X = 3N^{\frac{1}{2} - 2\beta}$ so that $\gamma = \frac{1}{2} - 2\beta$ in (4), it follows that

$$\delta < \frac{1}{6}\alpha - \frac{1}{2}\gamma + \frac{1}{4} = \frac{1}{6}\alpha + \beta.$$

This completes the proof. $\square$

For comparison, Sun et al. [24] derived, via a weaker Boneh-Durfee's attack, the bound on $d < N^{\delta_0}$:

$$\delta_0 < \frac{2}{3}\eta + \frac{5}{6} - \frac{4}{3}\sqrt{\eta^2 + \left(\frac{3}{2}\alpha - \frac{1}{2}\right)\eta - \frac{6\alpha - 1}{16}}.$$

Under our notation with $\eta = \frac{1}{2} - \beta$, this bound becomes

$$\delta_0 < \frac{7}{6} - \frac{2}{3}\beta - \frac{4}{3}\sqrt{\beta^2 - \left(\frac{3}{2}\alpha + \frac{1}{2}\right)\beta + \frac{6\alpha + 1}{16}}.$$

In contrast, our new bound is

$$\delta_0 = \delta + \frac{3}{4} - \frac{\alpha}{2} < \beta - \frac{1}{3}\alpha + \frac{3}{4}.$$

After some tedious computation, we find that our bound is demonstrably better for $\beta < \frac{1}{4} + \frac{2\sqrt{10}-7}{9}\alpha$.

Remark 2. It is interesting to observe that the attack bounds in Theorems 7 and 8 are indeed consistent. In Theorem 7 (with $1/4 < \beta_1 \leq 1/2$), the bound is $\delta < \frac{1}{6}\alpha - \beta_1 + \frac{1}{2}$, whereas in Theorem 8 (with $\beta_2 < 1/4$) the bound is $\delta < \frac{1}{6}\alpha + \beta_2$. Omitting identical parts in two bounds, i.e., $\frac{1}{6}\alpha$, the remaining ranges $\frac{1}{2} - \beta_1$ and β_2 coincide. It implies that Theorems 7 and 8 shall achieve the same best attack performance.

5 Experimental Results

To validate the correctness and effectiveness of our proposed attacks, we mainly conducted a series of numerical experiments based on Theorem 6 (since the other theorems derive from this main result). All experiments were executed on a machine running a 64-bit Windows 10 operating system with Ubuntu 22.04 installed via WSL 2. The implementation was performed in SageMath [29], and the parameters for generating experimental instances were chosen at random.

To generate simulated RSA instances, we first randomly selected two prime numbers p, q of a predetermined size and computed $N = pq$. Next, we generated a private exponent d of a fixed size δ_0 at random and computed its modular inverse e . Subsequently, we determined the most significant bits of $p + q$ to a prescribed ratio γ . Finally, we released the public values N , e , and S , with the known parameters $\log_2 N$, α , δ , γ , and δ_0 . The open-source implementation of the proposed attacks is available at https://github.com/MengceZheng/RSA_CFL.

During the experiments, we selected the parameter m and used an optimized t to control the lattice settings and perform the proposed attacks. Each experimental instance was repeated several times to ensure consistent extraction of the desired integer root. We collected a sufficient number of integer polynomial equations that satisfied the solvable condition, and then successfully extracted the target integer root. However, we observed that the experimental results deviate somewhat from the theoretical bound. To illustrate our proposed attacks, we present experimental results in Table 1 along with a numerical example.

Table 1. Experimental results of our new attack based on Theorem 6

$\log_2 N$	$\alpha = \log_2 e$	$\delta = \log_2 u , v $	$\gamma = \log_2 p + q - S $	$\delta_0 = \log_2 d$	m	t	ω	Time (s)
1024	0.999	0.051	0.426	0.301	3	8	158	59.437
1024	1.000	0.061	0.415	0.309	3	7	142	40.934
1024	0.998	0.072	0.393	0.319	3	6	126	26.882
1024	0.991	0.076	0.379	0.329	3	5	110	16.383
1024	0.998	0.101	0.361	0.350	4	5	180	236.091

Example 1. In this example, we consider an attack scenario where $\alpha \approx 1$, $\delta \approx 0.07$ and $\gamma \approx 0.4$. The RSA modulus N is constructed from two 256-bit primes p and q , resulting in a composite with $\log_2 N = 512$. The specific values for this RSA instance are as follows:

$$\begin{aligned}
N &= 667881289802154038823628680614836895352147070157135359517370 \backslash \\
&\quad 786146585569663805767598976469948361102331675600617530514952 \backslash \\
&\quad 0572078488495000186844811399641203, \\
e &= 172324352961620028297389245608298028274463258941568240927392 \backslash \\
&\quad 475948736506291258628505383961459191164157282917843848198210 \backslash \\
&\quad 2196501882009947221423361429104051, \\
S &= 166371334020806181832789298204468315927873129868852419579261 \backslash \\
&\quad 224686653643161600.
\end{aligned}$$

In order to execute the proposed attack, we search for the parameters $p_r, q_r, p_{r-1}, q_{r-1}$ and a_1, a_2, a_3, a_4 specified in Theorem 6. Then we set $m = 3$ and choose an optimized $t = 6$, which results in the construction of a 126-dimensional lattice. After approximately 8 seconds of computation, we successfully extract the desired root (x_0, y_0, z_0) . The extracted values are:

$$\begin{aligned}
x_0 &= 1202280999166873749666507784114830440272721682136332065817476, \\
y_0 &= 24562461685, \\
z_0 &= 8887884439.
\end{aligned}$$

Using the recovered value x_0 along with S , we compute $p + q = x_0 + S$, which in turn enables the factorization of $N = pq$ as follows:

$$\begin{aligned}
p &= 987112039719424516017990219085220105075795285600185553619734 \backslash \\
&\quad 35865055813332459, \\
q &= 676601300488637314332712754628200550868013854236643044900094 \backslash \\
&\quad 70957929895646617.
\end{aligned}$$

One may verify that $N = pq$, confirming the success of our proposed attack. Moreover, upon computing the private exponent d , we find that $d \approx N^{0.317}$,

which is an improvement over the bound achieved by previous Boneh-Durfee’s attack.

6 Concluding Remarks

In this work, we have presented a novel approach that combines continued fractions with Coppersmith’s lattice-based techniques to attack RSA with small private exponents. Our improved attacks use additional information such as partial leakage of prime factors or approximations of $p + q$ to derive higher bounds on the private exponent d . Specifically, by integrating the relation obtained from continued fraction of $\frac{e}{N}$ into a refined lattice-based attacks, we have established improved theoretical bounds, as demonstrated in Theorems 6, 7, 8. Numerical experiments further validated the feasibility and effectiveness of our approach. Besides, our method successfully factors RSA modulus with a larger private exponent, outperforming the previous attacks.

However, our experimental results indicate that there exists a non-negligible gap between the theoretical predictions (e.g., experimental $\delta \approx 0.07$ versus theoretical $\delta < 0.21$ in Example 1). This discrepancy suggests that there may be additional, yet unexplored, structural properties within our new attack that could be harnessed to further optimize the attack conditions. As a direction for future work, it is of interest to explore the hidden structures in the proposed attack, which may lead to even tighter bounds.

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